

1) Title: A Robust Speech Emotion Recognition (SER) Using Frozen Self-Supervised Speech Representations.

2) Student Information:

Sajjad Ibrahim
540861035
SajjadIbrahim123
<http://SajjadIbrahim123.github.io/elec5305-project-540861035>

3) Project Overview:

The project aims to investigate the robustness of speech emotion recognition (SER) systems using two self-supervised speech models, UniSpeech-SAT and WavLM. Currently, machine-human interfaces are capable of understanding spoken commands and performing complex tasks, however there is still a limitation in how well they can recognise emotional information from speech, especially when the speaker is unfamiliar or background noise is present. This project attempts to investigate this limitation by comparing how well both models represent emotional features when used as frozen feature extractors with the same lightweight MLP classifier. The models will be tested using actor-independent RAVDESS data and later under controlled environmental noise, allowing us to compare their performance on unseen speakers and determine how their different pretraining objectives affect robustness.

4) Background and Motivation:

There has been rapid growth in large language models and intelligent human-machine interfaces in the current decade, however there is still a clear gap in the ability of machines to interpret the emotional information contained within human speech. Human communication relies heavily on prosodic features such as pitch, speech rate, intensity and spectral characteristics, which provide important context beyond the literal words being spoken. Modern self-supervised speech models such as UniSpeech-SAT and WavLM are able to learn rich speech representations from large amounts of unlabelled audio, however their ability to generalise across unfamiliar speakers and noisy environments can differ depending on how they were pretrained. This project was selected to investigate these differences by using both models as frozen feature extractors with the same lightweight MLP classifier. By testing both systems on unseen RAVDESS speakers and later under controlled environmental noise, the project aims to determine whether their different pretraining objectives lead to different robustness characteristics for speech emotion recognition.

5) Proposed Methodology:

This project uses Python alongside Librosa, PyTorch and pretrained self-supervised speech models to build and compare two speech emotion recognition systems. The dataset used is a selected subset of the RAVDESS database containing neutral, happy, sad and angry speech recordings. Using Librosa, the raw .wav files are loaded and resampled to 16 kHz before being passed separately through UniSpeech-SAT and WavLM, which are kept frozen and used only as feature extractors. The frame-level speech representations produced by each model are reduced into 768-dimensional embeddings using temporal mean pooling. These embeddings are then passed into the same lightweight Multi-Layer Perceptron (MLP) classifier so that the comparison between both models remains controlled.

The dataset is divided using actor-independent training, validation and test splits to measure performance on unseen speakers. Environmental noise from the DEMAND dataset will also be added at controlled SNR levels to compare how well each model maintains emotion recognition performance under noisy conditions.

Week	Task
1 - 2	Chose Speech Emotion Recognition (SER) as the overall project topic, with the original idea focusing on a real-time emotion recognition system using MFCC features, an MLP classifier and live microphone input.
3 - 5	Investigated existing SER approaches, selected the RAVDESS dataset, explored Librosa and PyTorch, and began planning audio preprocessing, MFCC extraction and noise augmentation methods.
6 - 7	Continued researching speech emotion recognition methods, relevant datasets, feature extraction approaches and possible deep learning architectures while refining the overall project direction.
8	Following feedback from Professor Craig, the project was changed from developing a general real-time SER application to a more focused research comparison between UniSpeech-SAT Base+ and WavLM Base+. The new research question investigates whether differences in their pretraining objectives produce different robustness to unseen speakers and acoustic noise when used as frozen feature extractors.
8 - 9	Filtered RAVDESS to neutral, happy, sad and angry recordings, resampled audio to 16 kHz, created actor-independent train/validation/test splits, extracted 768-dimensional embeddings using frozen UniSpeech-SAT and WavLM, and developed the same lightweight MLP classifier for both systems.
9 - 10	Clean Unseen-Speaker Evaluation: Trained both MLP classifiers and evaluated them using completely unseen RAVDESS actors. Measured Macro-F1, per-emotion F1 and confusion matrices, producing initial clean test results of 0.7573 for UniSpeech-SAT and 0.8186 for WavLM.
10 - 11	Acoustic Noise Robustness Evaluation: Introduce real environmental noise from the DEMAND dataset at controlled 10 dB and 0 dB SNR levels and compare how much the performance of UniSpeech-SAT and WavLM decreases relative to the clean baseline.
12 - 13	Analyse the complete experimental results, compare unseen-speaker and noise robustness, finalise the research conclusions, complete the final report, update the public GitHub repository.

Ai Acknowledgement: Ai was used to aid in the writing of the code of the assignment.

References:

- 1) Chen, S., Wu, Y., Wang, C., Chen, Z., Chen, Z., Liu, S., Wu, J., Qian, Y., Wei, F., Li, J., & Yu, X. (2022). *UniSpeech-SAT: Universal speech representation learning with speaker aware pre-training*. 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 6152–6156.
- 2) Chen, S., et al. (2022). *WavLM: Large-scale self-supervised pre-training for full stack speech processing*. IEEE Journal of Selected Topics in Signal Processing, 16(6), 1505–1518.
- 3) Livingstone, S. R., & Russo, F. A. (2018). *The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in North American English*. PLOS ONE, 13(5), e0196391.
- 4) Atmaja, B. T., & Sasou, A. (2022). *Evaluating self-supervised speech representations for speech emotion recognition*. IEEE Access, 10, 124396–124407.
- 5) Pepino, L., Riera, P., & Ferrer, L. (2021). *Emotion recognition from speech using wav2vec 2.0 embeddings*. *Proceedings of Interspeech 2021*.